

University Paris 1 Panthéon-Sorbonne

Master 2 TIDE — Traitement de l'Information et Data Science en Entreprise

High Dimensional Statistics

Sustainable Energy Analysis

Global Sustainable Energy Data (2000–2020)

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List of Acronyms

SDG	Sustainable Development Goal
ML	Machine Learning
HDI	Human Development Index
LR	Linear Regression
VIF	Variance Inflation Factor
RMSE	Root Mean Squared Error

Introduction

Contributions

- **Anne-Elisabeth Makosso** (Q1): Predicting Primary energy consumption per capita and the factors that drive it.
- **Amine Sellami** Data Preprocessing & (Q2): Predicting access to clean fuels for cooking.
- **Alexis Mattei** (Q3): Identification of country energy profiles via PCA and K-Means clustering, and temporal evolution analysis (2000–2020).

Data Preprocessing

The dataset contains 3649 examples, from 276 different countries, and 21 columns. Most of the countries have records from 2000, representing 21 observations per country. However, we have some data points that start after that year for some other entities.

In the provided dataset, there are some columns that exceed the 50% of missing values, like *Financial flows to developing countries (US \$)*, as shown in Figure 1. From our analysis, the column *Renewables (% equivalent primary energy)* is either present for a given country or it is totally absent. Its 59.71% of missing values indicates that the observations are absent for 59.71% of the entities. As a result, we will drop this column from our dataset. Moreover, there are approximately 5 countries that have missing values in *Access to clean fuels for cooking*. Instead of doing imputation, which may introduce false values, we are going to drop these examples. We go from 3649 to 3375 examples, which is sufficient for an ML algorithm.

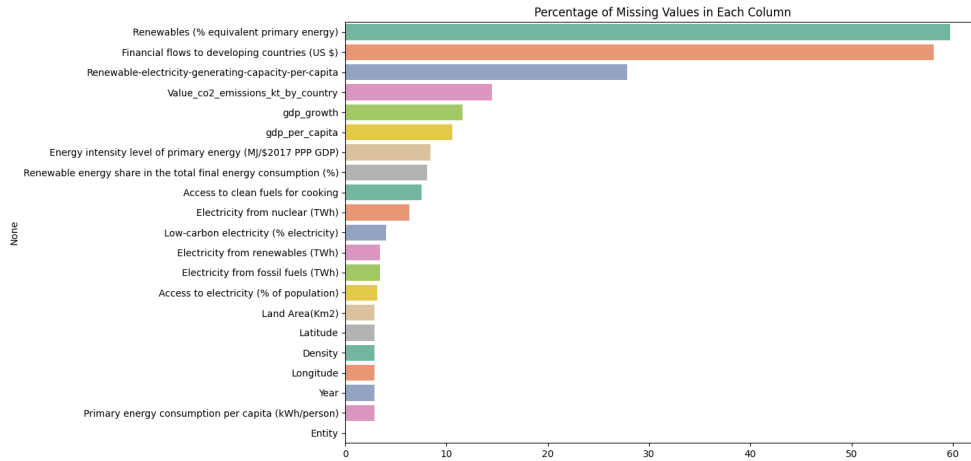


Figure 1: Percentage of missing values

To deal with the other missing values, and to add additional information to the selected models, we introduce a new variable named *Continent*. In fact, after grouping the number of missing values in the *Financial flows to developing countries (US \$)* column, thanks to the code in the Appendix, we notice that the European countries have 100% missing values. This makes sense because this indicator measures the aid received by developing countries for clean-energy projects. According to World Population Review, a country that has HDI between 1 and 0.8 is considered a developed country. So, we design a dictionary containing all the developed countries that have missing values in that column, and we fill it with 0. For the rest of the countries, that have at least one observation, we fill with the median as the safest approach, as this column has some outliers.

The access to electricity has missing values, namely in two countries: Liberia and Guinea-Bissau. The minimum value in Guinea-Bissau is recorded at the third observation with 1.25% and for Liberia it is 1.27%. As we believe it is impossible to have 0% of access, we fill the missing values with 1%. In addition, according to Figure 9, there is a perfect linear relationship between *Value_co2_emissions_kt_by_country* and *Electricity from fossil fuels (TWh)*, as they have a Pearson coefficient equal to 0.99. Thus, we train

an **LR** model, and then we impute the missing values with the obtained intercept and slope coefficients. The R^2 coefficient confirms our intuition, since we obtained $R^2 = 0.9796$. Since for a given row, only one of them is missing and not both, we can perform the imputation thanks to the obtained relationship. On the other hand, according to **International Atomic Energy Agency (IAEA) - PRIS Database**, Chile, Indonesia, Kazakhstan, Malaysia, Saudi Arabia, and Tuvalu do not produce nuclear electricity. And in particular, despite Kazakhstan being a major uranium producer, it has had no operating nuclear power plant since the BN-350 shut down in 1999. As a result, from 2000 they have 0 TWh. Moreover, Tuvalu also has 100% missing values in *Electricity from renewables (TWh)* and *Low-carbon electricity (% electricity)*. To impute these columns, we develop a KMeans clustering algorithm based on the latitude and longitude of Oceania countries. Based on Silhouette Score and Inertia Score, as shown in Figure 10, the best K is 6. The obtained clusters are shown in Figure 2.

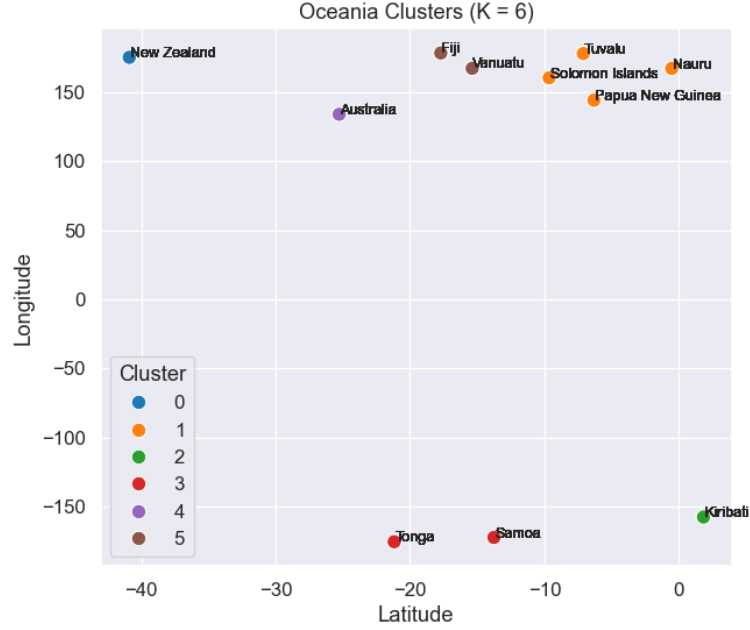


Figure 2: Oceania Clusters

Table 1: Statistical Results for Oceania Clusters - Electricity from renewables (TWh)

Cluster	mean	std	min	25%	50%	75%	max
0	31.28	3.56	24.84	28.07	31.4	34.7	36.3
1	0.42	0.61	0	0	0	1.16	1.49
2	0	0	0	0	0	0	0
3	0.026	0.026	0	0	0.02	0.05	0.07
4	28.88	13.14	17.11	18.5	21.19	36.15	63.99
5	0.28	0.28	0	0.0025	0.215	0.5575	0.71

According to KMeans clustering, Tuvalu is assigned to the first cluster which has a skewed distribution. We will fill the missing values of the entire country with 0. The values in the table suggest that they represent small island nations or small economies with tiny renewable electricity production. Furthermore, we do the same process for *Low-carbon electricity (% electricity)*, and we have the same value to impute for Tuvalu.

Besides that, for *Renewable energy share in the total final energy consumption (%)*, it has only one missing value per country, which is the last registered year. Albania has more than one missing value, so we will use the forward fill technique. If for some country, it is the first value, we fill it backward. It is 100% absent in Albania, so we will repeat the same process for what we did with Energy columns. Albania is considered in the cluster 0 according to Figure 3. This cluster has a symmetric distribution since the mean value is 30.1 aligned with the median 30.29, with a very low standard deviation value of 1.35. Hence, we fill the missing values in Albania with the mean.

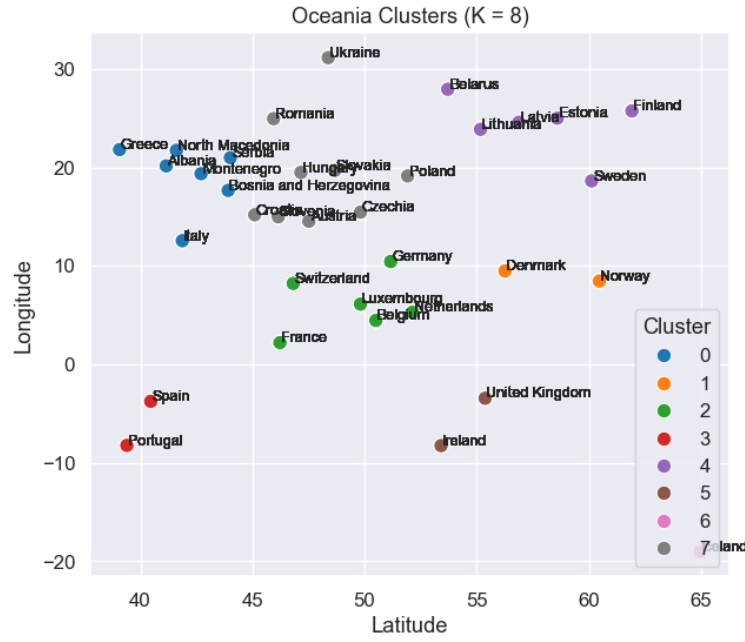


Figure 3: Europe Clusters

Finally, for the *GDP Growth*, we have 100% missing values in 11 countries and for the others it is totally present. We see that dropping these countries from our analysis is more suitable to our model, letting us have in the end 3144 examples, which is sufficient for an ML model.

1. Q1 — Predicting Primary energy consumption per capita and the factors that drive it

Contribution: Anne-Elisabeth Makosso

1.1 Research Question and Motivation

Primary energy represents "all energy products not transformed, directly exploited or imported [...] such as crude oil, oil shale, natural gas, solid mineral fuels,...(INSEE)". Primary energy consumption "measures the total energy demand of a country and covers consumption of the energy sector itself, losses during transformation and distribution of energy, and the final consumption by end users (Eurostat)". Dividing it per capita allows to compare countries. So, this variable reflects the level of development of a country, its resources and the intensity of energy use within a country. And all these effects can sometimes be contradictory as can be explained by the resource curse theory. Predicting this variable can help to predict future energy demand and, in particular understand the factors that explain primary energy consumption.

In this section, we want to predict the evolution of Primary energy consumption per capita (kWh/person) using machine learning methods but also identify the variables that impact primary energy consumption per capita.

1.2 Methodology

We started by creating groups of countries based on the evolution of Primary energy consumption per capita (kWh/person) for each country using KMeans. Four groups were found using the method of inertia and silhouette :

- Group 0 : developing countries, with a slight increase over the years (Afghanistan, Angola, Burkina Faso, Ethiopia, Niger, Rwanda,...)
- Group 1 : Very rich countries with some Gulf states that are rich in raw materials. Slight decrease recently (Canada, Iceland, Kuwait, Norway, Qatar, Singapore, United Arab Emirates)
- Group 2 : Industrialized countries with high consumption (Australia, France, Sweden)
- Group 3 : Often emerging countries or developed countries but not very consumers (China, South

Africa, Mexico, Italy, Spain)

Even though it won't necessarily be a useful variable, it can be interesting to understand the profiles of the countries we are studying.

To predict the variable at time t , explanatory continuous variables were lagged by one period $t - 1$. Also, based on the methodology proposed by *Martyushev, Nikita V et al.*, two other variables were created for each country : the trend (using the OLS method) and its residuals for each country. These two variables aim to understand the dynamics of the target through time.

Once all the variables were created, we have to check if there are any multicollinearities and also remove some variables.

First, we have decided to remove geographical variables such as longitude, latitude or density because only by using these variables, unique for each country, the model could learn which country it is and therefore learn what happened in previous years to predict the new variable.

Then, using the VIF method, we have removed variables such as *Lag1_Access to electricity (% of population)*, *Lag1_Access to clean fuels for cooking* and *Lag1_Electricity from fossil fuels (TWh)*.

Concerning categorical variables, neither Label nor Continent were kept. Actually, Cramer's V statistic has shown that there is a strong association of these two variables with the target (very close to 1).

For explainability and comparison reasons, all continuous explanatory variables were standardized.

For the modeling step, we have started from the simplest to the most complex and better performing models.

First, we estimated a simple regression using OLS to understand the relationships between variables. Even though all variables were statistically significant at the 5% level, some coefficients were close to zero. To select more pertinent variables, a LASSO regression was implemented. The alpha parameter was selected using the LASSO path and it allowed us to remove one variable, *Lag1_gdp_growth*.

This model was good to select variables but we thought that the performance could be improved. In order to do that, we implemented a very basic XGBoost Regressor model using the variables selected by LASSO.

1.3 Results

Although there is a slight overfitting between the train and the test, we can still see a strong predictive performance and good results from the XGBoost Regressor model. This can be explained because there are many developing countries with low values, therefore the gap between the real values and the prediction is small but this gap is higher for the richest countries. This hypothesis is quite verified with the plot of residuals which reveals the presence of heteroskedasticity.

Table 2: XGBRegressor Results

	Evaluation	Mean_CV_Train	Test
0	MSE	0.020572	0.029096
1	RMSE	0.141571	0.170575

However, the simple results can't be interpreted.

1.4 Conclusion

In order to have a better interpretability and understand which explanatory variables are the most influent, we used SHAP values. Therefore, we can conclude that :

- the level of development of a country through GDP per capita is the principal variable that contributes to predict the primary energy consumption per capita in our model
- the energy intensity is the second variable that contributes the most to the prediction which is logical because countries which produce energy inefficiently will probably have a higher primary energy consumption per capita for example
- the share of renewable energy in the total final energy consumption is the third variable that contributes the most to predicting our target. This could be explained because, for instance, rich

- countries that consume a lot of energy tend to invest more in renewable energy
- financial flows contribute to predicting primary energy consumption maybe because developed countries already have a high primary consumption and on the other hand developing countries take advantage of this financial aid.
- the contribution of the residuals demonstrates that the gap around the trend can provide useful information to predict primary energy consumption
- the contribution of the other variables to predict our target is very negligible, if non-existent.

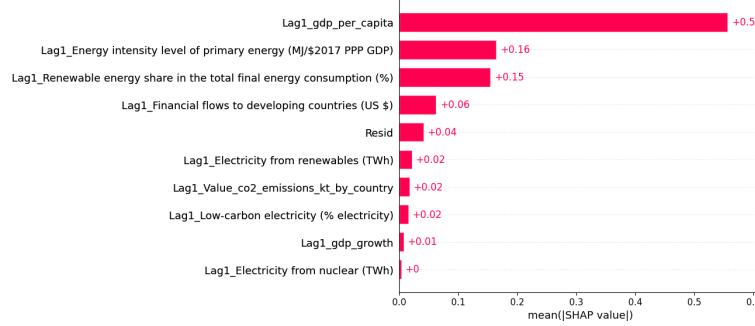


Figure 4: SHAP values for XGBoost

2. Q2 — Predicting access to clean fuels for cooking

Contribution: Amine Sellami

2.1 Research Question and Motivation

Motivated by the SDG 7, which is **Affordable and Clean Energy**, it is essential to look for the probability that a person can have access to clean fuels for cooking. In fact, while electricity access and renewable energy production tend to dominate policy discussions, cooking fuels directly affect health and environmental sustainability.

In this section, we employ the *XGBoostRegressor* as a model to predict access to clean fuels for cooking, aligning with *GridSearchCV* to select the appropriate parameters in order to minimize the *RMSE*.

2.2 Methodology

In this part, we start by adding the statistical characteristics of access to clean fuels for cooking grouped by continent, as discussed in the previous section. In addition, we employed VIF in order to reduce the multicollinearity. As a result, we will keep only the mean characteristic from the created features, and we will drop the *Electricity from nuclear (TWh)* and *Value_co2_emissions_kt_by_country* because they have a very high VIF value, more than 70.

To start, we split the dataset into two training datasets and a testing dataset using *train_test_split*. The first one is used to train our model and the second one is used to test our selected model on unseen data. For the categorical columns, we keep only the *Continent* that we encoded using *LabelEncoder*, and we drop *Entity*. In fact, encoding the *Entity* column will create many variables, and it will be hard to the model to interpret, and we can detect a country also by its *Latitude* and *Longitude*. Furthermore, we can clearly see from [Figure 5](#) that the probability of having access to clean fuels for cooking is different from continent to continent. In fact, in average, 94.1% of the population of the European countries have access to clean fuels, which is very far from the African countries that have only 24.8% in average. The latter value is also far from the 50% of the countries that have only 6.8%. This suggests that in one continent, we can have countries currently under development and others that are far away.

Moreover, we adopted *GridSearchCV* to select the appropriate hyperparameters. As our data samples are around 2000 after split, we need to ensure low values for *max_depth* and *n_estimators* to prevent overfitting, which justifies our choice of the different values in [C.1](#). Since we are dealing with percentages, we use RMSE as our metric.

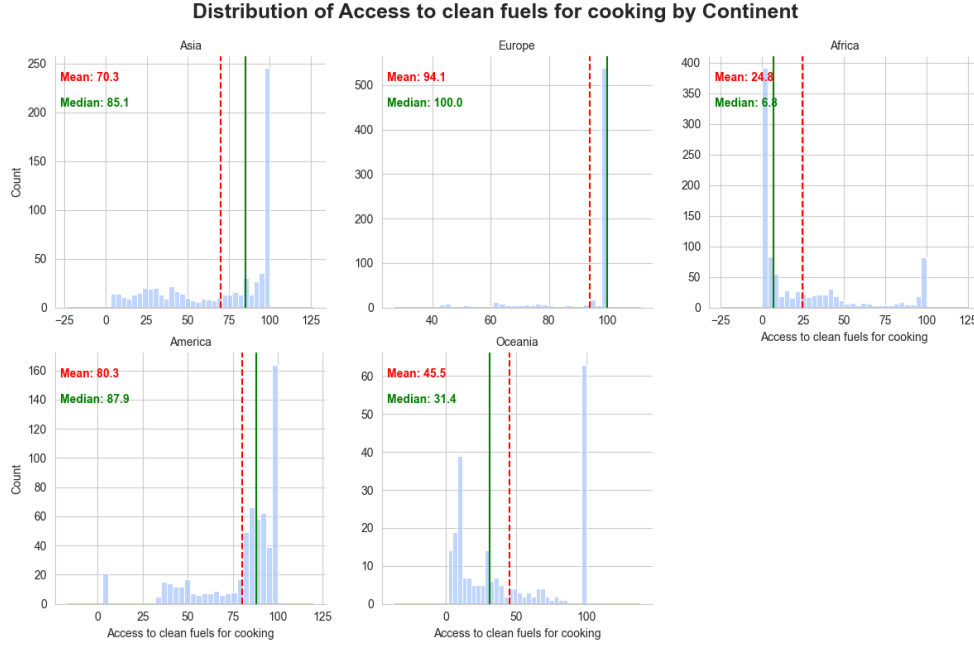


Figure 5: Distribution of Access to clean fuels for cooking by Continent

2.3 Results

GridSearchCV selected: *colsample_bytree* = 0.6, *learning_rate* = 0.01, *max_depth* 6, *n_estimators* = 300, *subsample* = 0.7. Furthermore, we employ *KFold* technique, setting the number of folds as 5, as developed in C.1. Table 3 summarizes the training results.

Table 3: XGBRegressor Results

Fold	Train	Validation
1	3.435	4.351
2	3.410	4.485
3	3.387	4.828
4	3.392	4.179
5	3.388	4.204

The mean training RMSE is equal to 3.4027, the mean validation RMSE is equal to 4.409. We employ then our prediction to the unseen data, and we obtain a test RMSE of 4.41. These results prove the good quality of our model, preventing overfitting.

3. Q3 — Identifying Country Energy Profiles via PCA and Clustering

Contribution: Alexis Mattei

3.1 Research Question and Motivation

The question I chose to investigate is the following: can we identify distinct country profiles in terms of energy transition using PCA, and how have these profiles evolved between 2000 and 2020?

We know that we have 151 countries observed over up to 21 years and 12 energy-related variables, so a direct comparison across all dimensions is simply not readable. PCA felt like the natural starting point because it lets me reduce the dimensionality of the data while keeping the main sources of variation, and project countries into a space where they can actually be compared visually. I then complemented this with K-Means clustering to group countries sharing similar energy characteristics, and use the temporal dimension of the dataset to track how these groups have shifted over the study period.

3.2 Methodology

Variable selection. I selected 12 variables covering energy access, production mix, consumption, emissions and economic development: access to electricity, access to clean fuels for cooking, renewable

energy share, electricity from fossil fuels, nuclear and renewables, low-carbon electricity share, primary energy consumption per capita, energy intensity, CO₂ emissions, GDP per capita, and financial flows to developing countries. The variable *Renewable-electricity-generating-capacity-per-capita* was excluded due to 826 missing values (26% of observations) — keeping it would have forced me to drop too many country-year pairs.

Standardization. The variables are expressed in very different units (TWh, %, USD, kt...), which is why I standardized each one to zero mean and unit variance before running PCA. This is indeed really important to do because variables with large absolute values like CO₂ emissions would mechanically dominate the first components, regardless of their actual informational content.

PCA. I fitted the PCA on the full dataset (all years pooled) rather than separately for 2000 and 2020. This gives a common projection space, which is essential to make temporal comparisons meaningful: a country’s position in 2000 and in 2020 can be read on the same axes. The scree plot is provided in [Appendix \(Q3, Scree Plot\)](#). The first two components explain 33.3% and 23.1% of total variance respectively (56.5% combined), and 5 components are needed to reach 80%, which is the threshold I chose for clustering. PC1 can be read as an *energy development level* axis: countries with high electricity access, GDP per capita and fossil consumption score positively, while countries with high renewable share score negatively. PC2 captures more of the *energy mix*: large CO₂ emitters and heavy fossil electricity producers score positively. We can indeed see that the correlation circle ([Appendix Q3, Correlation Circle](#)) provides a more detailed view of these loadings.

Clustering. I applied K-Means on the first 5 principal components instead of the raw data, to avoid redundant information from correlated features. I chose $K = 4$ because of both the elbow method (inflection point in inertia) and the silhouette score. I also tested Spectral Clustering with $K = 4$, but the result was unusable: one cluster contained 3,116 of the 3,144 observations, while the three others were essentially empty. This suggests the data has a convex structure, for which K-Means is better suited.

3.3 Results

Country projections: 2000 vs 2020. Figure 6 shows where countries land in the PC1–PC2 plane for each of the two years. What strikes me first is how stable the overall structure is — the main groups are already visible in 2000. That said, several countries have clearly moved along PC1, reflecting a rise in energy development level. The most striking shift is China’s: it moves from a moderate central position in 2000 to an extreme position along both axes in 2020, pushed far right by its industrial expansion and fossil fuel growth. India and Indonesia have also moved noticeably rightward, while European countries barely move — their energy profiles were already mature in 2000. The detailed trajectories for selected countries are shown in [Appendix \(Q3, Country Trajectories\)](#).

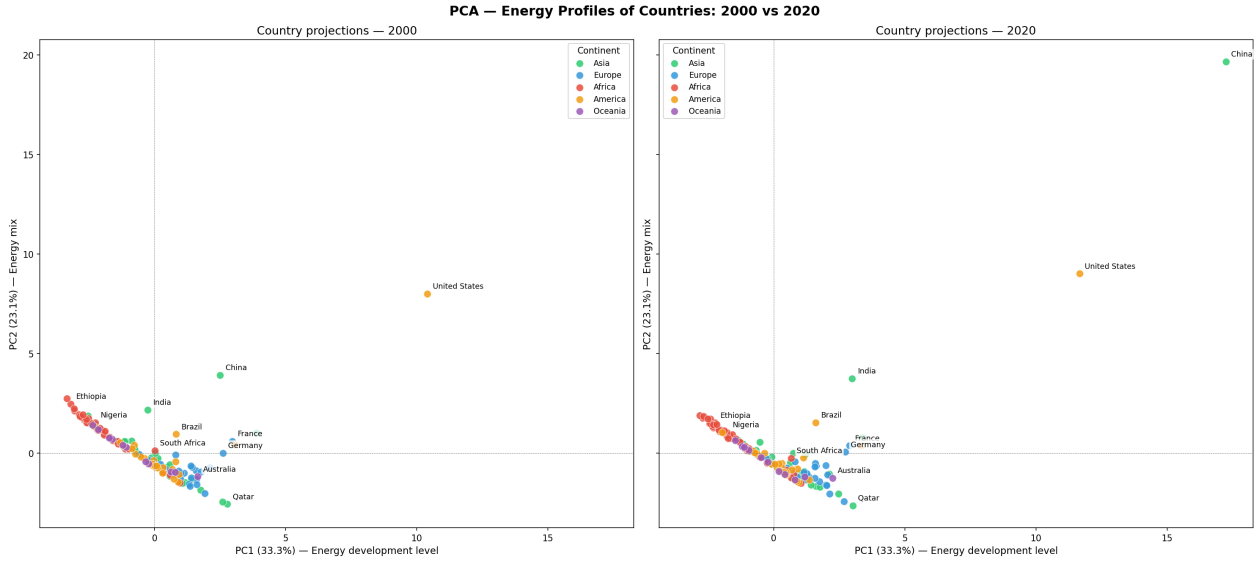


Figure 6: Country projections in the factorial plane (PC1 vs PC2) for 2000 (left) and 2020 (right). Each point represents one country, colored by continent.

Clustering results. Figure 7 presents the K-Means partition ($K = 4$) for 2000 and 2020. Four interpretable groups emerge:

- **Cluster 0 — Low-development / high renewables** (mainly Sub-Saharan Africa): low electricity access, low GDP, but paradoxically high renewable share. This is not a sign of a successful energy transition — these countries rely on biomass and hydropower by necessity, not by policy.
- **Cluster 1 — Fossil giants** (China, USA): off the charts in terms of fossil electricity, CO₂ emissions and primary energy consumption.
- **Cluster 2 — Rich and developed** (Western Europe, Qatar, Australia): high GDP, near-universal energy access, and comparatively lower energy intensity.
- **Cluster 3 — Emerging economies** (India, Brazil, South Africa, Indonesia): intermediate development, growing fossil consumption, but improving access to energy services.

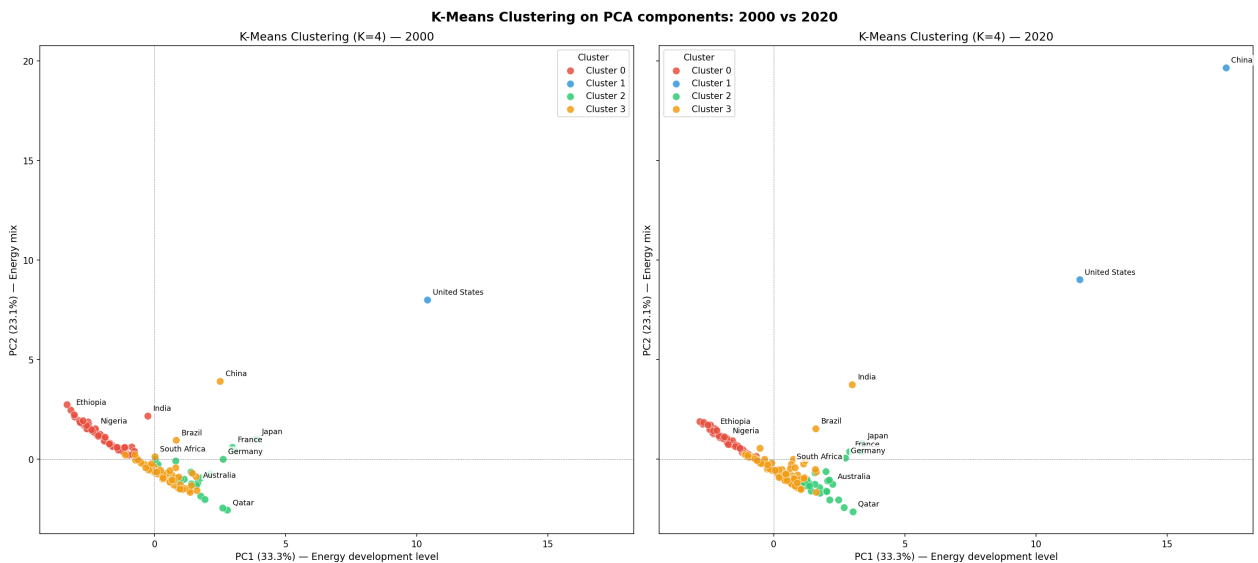


Figure 7: K-Means clustering ($K = 4$) on the first 5 PCA components for 2000 (left) and 2020 (right).

Between 2000 and 2020, we notice that **30 out of the 148 countries present in both 2000 and 2020 changed cluster (20%)**. The most significant transitions are: China (Cluster 3 $\rightarrow$ 1, joining

the fossil giants); India, Indonesia and Pakistan (Cluster 0 → 3, graduating to emerging economies); Denmark, Ireland and the Netherlands (Cluster 3 → 2, moving toward a richer and cleaner profile).

Cluster profiles. Figure 8 shows the standardized mean values per cluster. I used z-scores here to allow comparisons across variables with different units: a positive bar means the cluster is above the global average for that variable, a negative bar means below. The profiles confirm our interpretations: Cluster 0 stands out for its high renewable share and low everything else; Cluster 1 is extreme on all fossil and emissions indicators; Cluster 2 scores high on GDP and access; Cluster 3 sits in between on most variables, which makes it the most heterogeneous group.

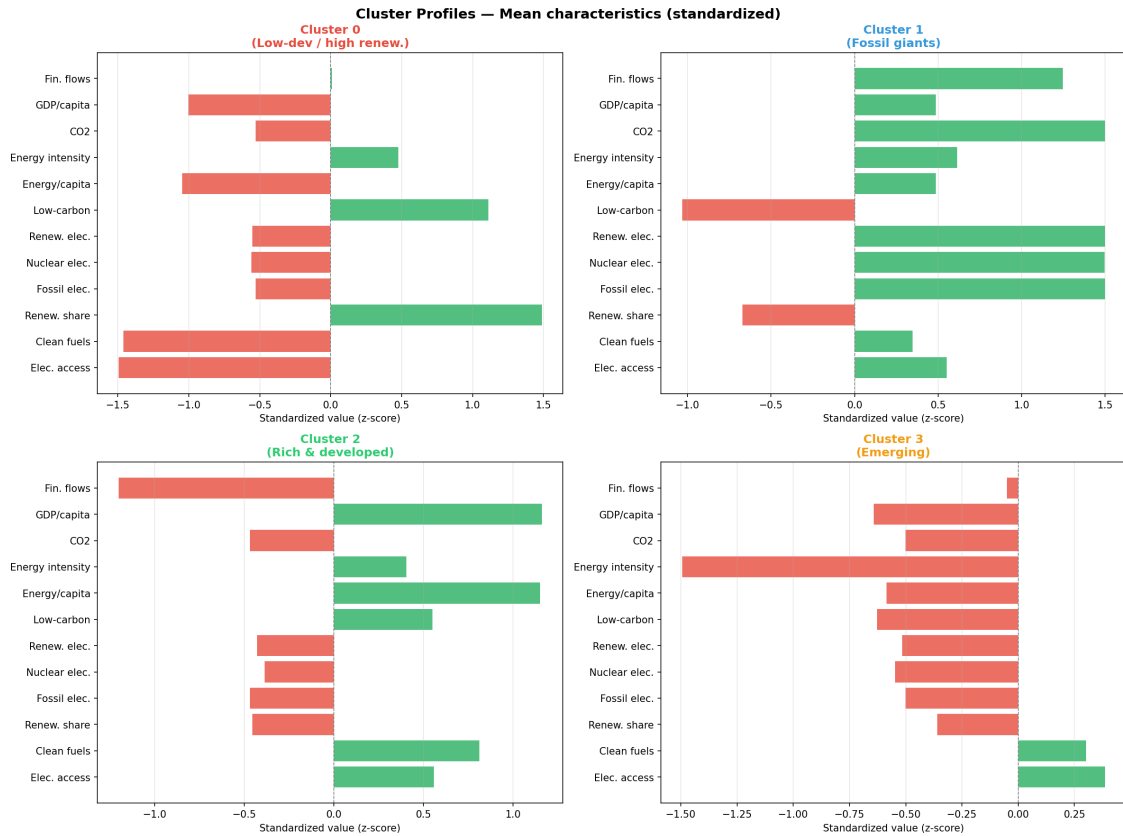


Figure 8: Standardized mean values (z-scores) per cluster. Green bars indicate above-average values, red bars below-average.

3.4 Conclusion

What this analysis shows, I think, is that the global energy landscape is structured and persistent. Two main axes — development level and energy mix — explain most of the variation between countries, and the four clusters identified by K-Means are interpretable and largely stable over time. The 20% of countries that did change cluster between 2000 and 2020 tell the more interesting story: China’s rise as a fossil giant, the emergence of India and Indonesia, and the quiet consolidation of European countries into a cleaner and wealthier group. The energy transition is real, but it is happening at very different speeds — and for some of the poorest countries in Cluster 0, what looks like a “green” profile is in fact just a reflection of energy poverty.

Appendix — Preprocessing

To ensure that a reader can understand how we created the code, we introduce in this appendix some of the functions that we used in our work.

A. Data Preprocessing

As mentioned earlier, we introduced a new column named *Continent* thanks to the *country_converter* library in python.

```
def get_continent(country_name):
    try:
        continent = coco.convert(names=country_name, to='continent')
        if continent == country_name or continent == 'not_found':
            return None
        return continent
    except:
        return None
```

To observe how many missing values we have for *Financial flows to developing countries (US \$)* column, we compute:

```
pd.crosstab(
    df['Continent'],
    df['Financial_flows_to_developing_countries_(US$)'].isna(),
    rownames=['Continent'],
    colnames=['Missing']
)
```

The heatmap representing the pearson correlation between the different numerical columns is given in [Figure 9](#) To select the adequate number of clusters, we compute the Inertia and Silhouette score. The best number is a tradeoff between these two coefficients. We need to minimize the Inertia score and maximize the Silhouette. In the context of grouping the Oceania countries, we have the following figures.

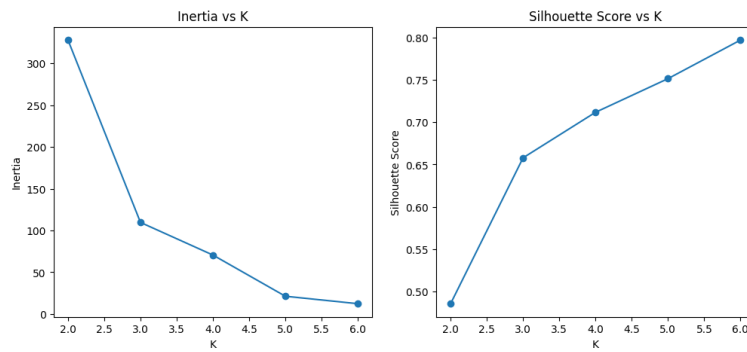


Figure 10: Inertia and Silhouette Score for Oceania Countries

For the Europe continent, the best number of clusters is 8, according to [Figure 11](#).

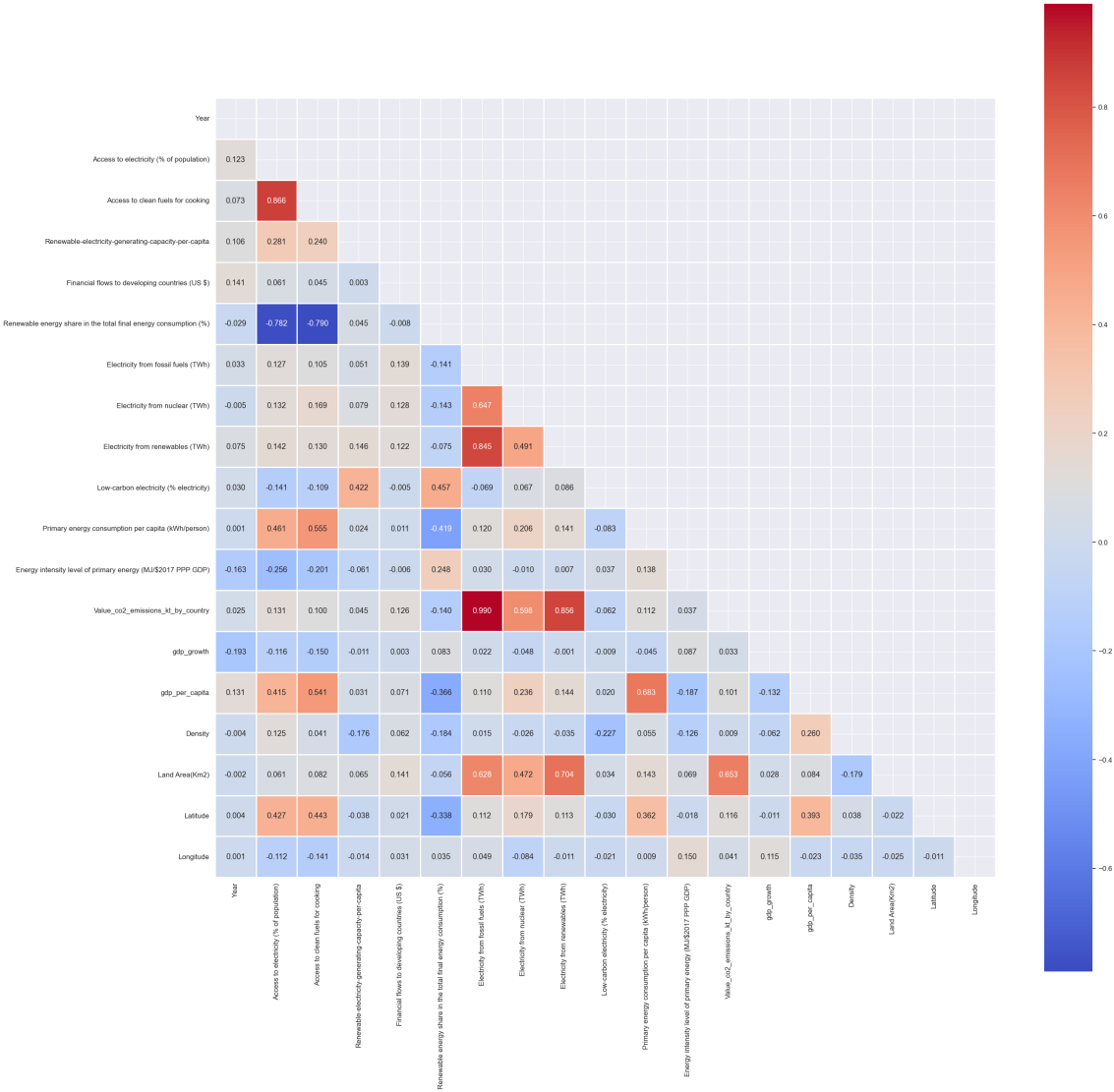


Figure 9: Heatmap - Correlation

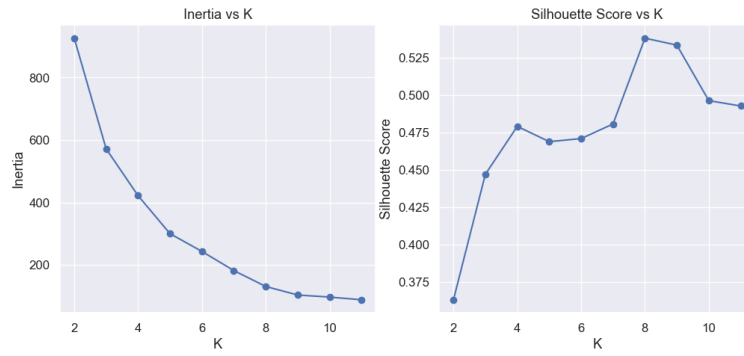


Figure 11: Inertia and Silhouette Score for European Countries

B. Feature Engineering

The function that is responsible to create new columns that calculate the mean, standard deviation, min, max, and skew based on a feature is given by:

```
def groupby_features(df, agglst, target):
    for col in agglst:
        df[f"{col}__mean"] = df.groupby(col)[target].transform('mean')
        df[f"{col}__std"] = df.groupby(col)[target].transform('std')
        df[f"{col}__min"] = df.groupby(col)[target].transform('min')
        df[f"{col}__max"] = df.groupby(col)[target].transform('max')
    return df
```

C. Modeling

C.1 XGBoost - Access to clean fuel for cooking

To employ *GridSearchCV* for *XGBRegressor*, we develop the following code:

```
param_grid = {
    'learning_rate': [0.001, 0.01],
    'max_depth': [2, 4, 6],
    'subsample': [0.6, 0.7],
    'colsample_bytree': [0.6, 0.7, 0.8],
    'n_estimators': [100, 200, 300]
}

model = XGBRegressor()
grid = GridSearchCV(model, param_grid, cv=5, scoring='
    neg_root_mean_squared_error', n_jobs=-1)
grid.fit(X_train, y_train)

print(grid.best_params_)
print(grid.best_score_)
```

For developing *KFold*, we employ this code:

```
# Hyperparameters for XGBRegressor
hyperparameters = {
    'learning_rate': 0.01,
    'max_depth': 6,
    'subsample': 0.7,
    'colsample_bytree': 0.6,
    'n_estimators': 300
}
```

```

model = XGBRegressor(**hyperparameters)

cv = KFold(n_splits=5, shuffle=True, random_state=42)

val_rmse = []
val_mse = []
train_rmse = []
train_mse = []

for i, (train_index, val_index) in enumerate(cv.split(X, y)):

    X_train, X_val = X.iloc[train_index], X.iloc[val_index]
    y_train, y_val = y.iloc[train_index], y.iloc[val_index]

    model.fit(X_train, y_train)

    train_preds = model.predict(X_train)
    val_preds = model.predict(X_val)

    # Metrics
    tr_mse = mean_squared_error(y_train, train_preds)
    tr_rmse = np.sqrt(tr_mse)

    v_mse = mean_squared_error(y_val, val_preds)
    v_rmse = np.sqrt(v_mse)

    train_rmse.append(tr_rmse)
    train_mse.append(tr_mse)
    val_rmse.append(v_rmse)
    val_mse.append(v_mse)

    print(f"\nFold_{i+1}")
    print("_____")
    print(f"Train_RMSE: {tr_rmse}")
    print(f"Validation_RMSE: {v_rmse}")

print("\n=====FINAL_RESULTS=====")
print(f"Mean_Training_RMSE: {np.mean(train_rmse)}")
print(f"Mean_Validation_RMSE: {np.mean(val_rmse)}")

```

Appendix — Q1

Kmeans on the target variable

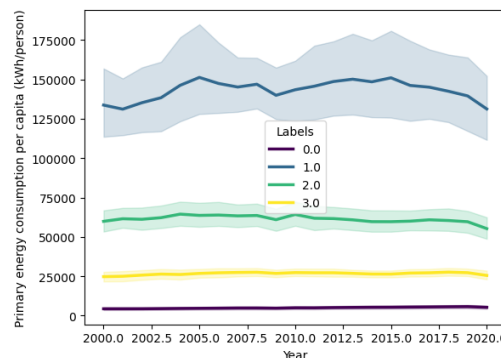


Figure 12: Average evolution of primary energy consumption per capita for each cluster

LASSO path to select optimal alpha

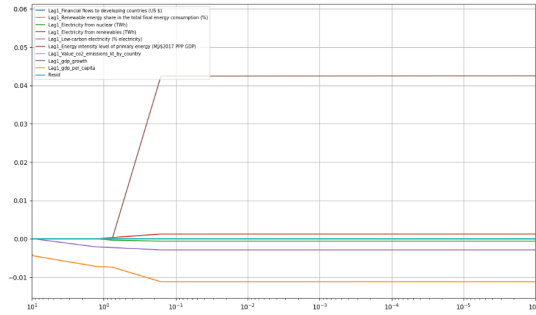


Figure 13: Select pertinent variables using LASSO regularization path

Model checking

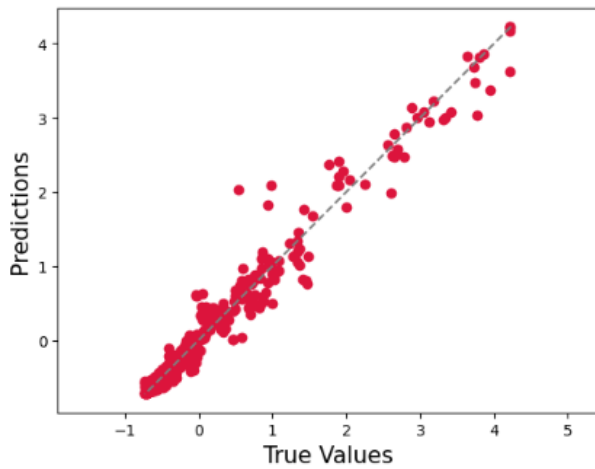


Figure 14: Predicted VS true values of primary energy consumption per capita

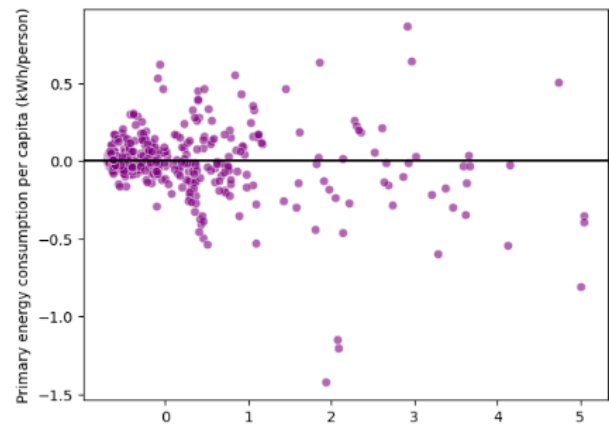


Figure 15: Residual plot. It shows the presence of heteroskedasticity

Appendix — Q3

Scree Plot

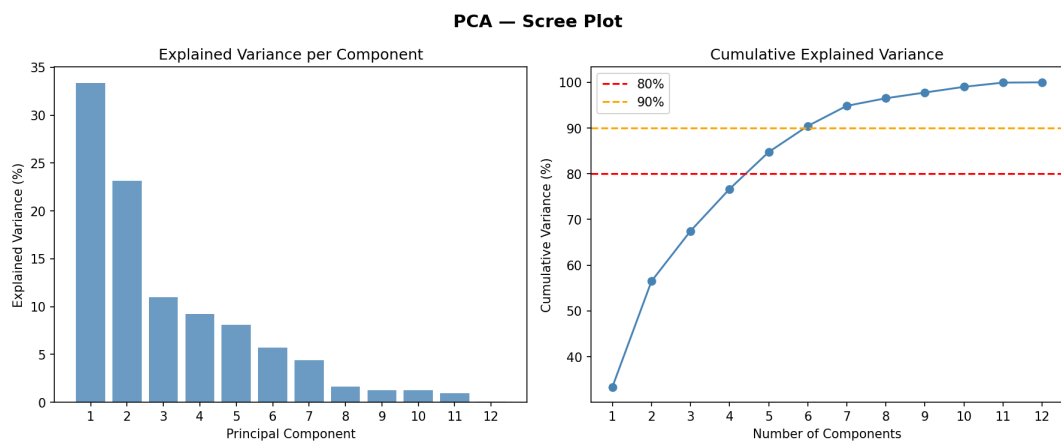


Figure 16: Scree plot: explained variance per component (left) and cumulative explained variance (right).

Correlation Circle

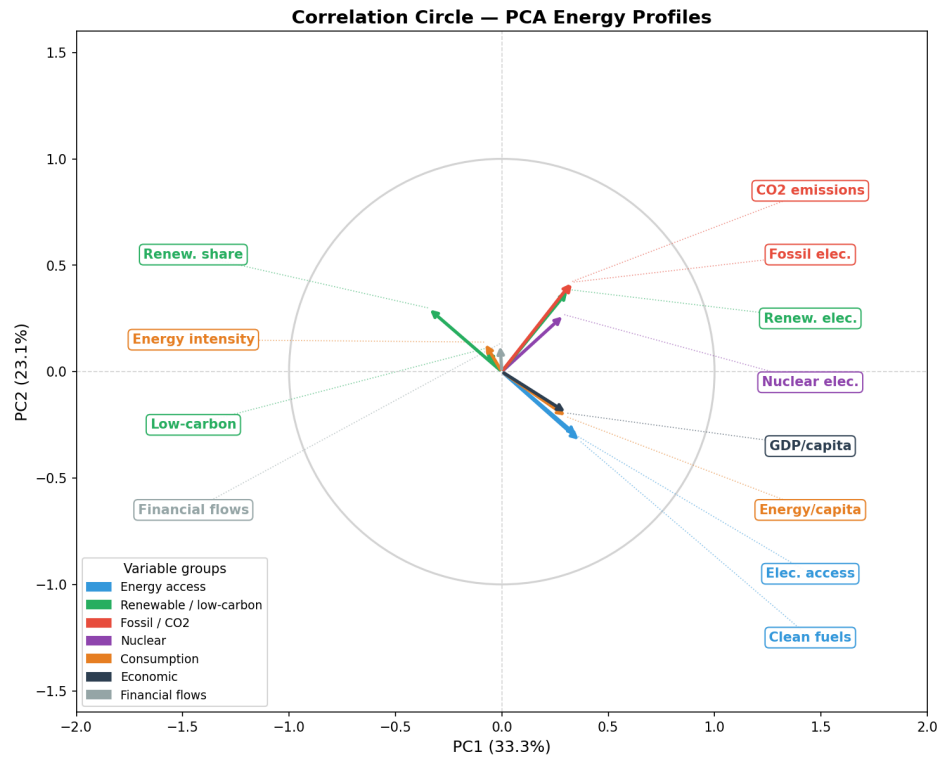


Figure 17: Correlation circle showing the contribution of each variable to PC1 and PC2.

Country Trajectories 2000–2020

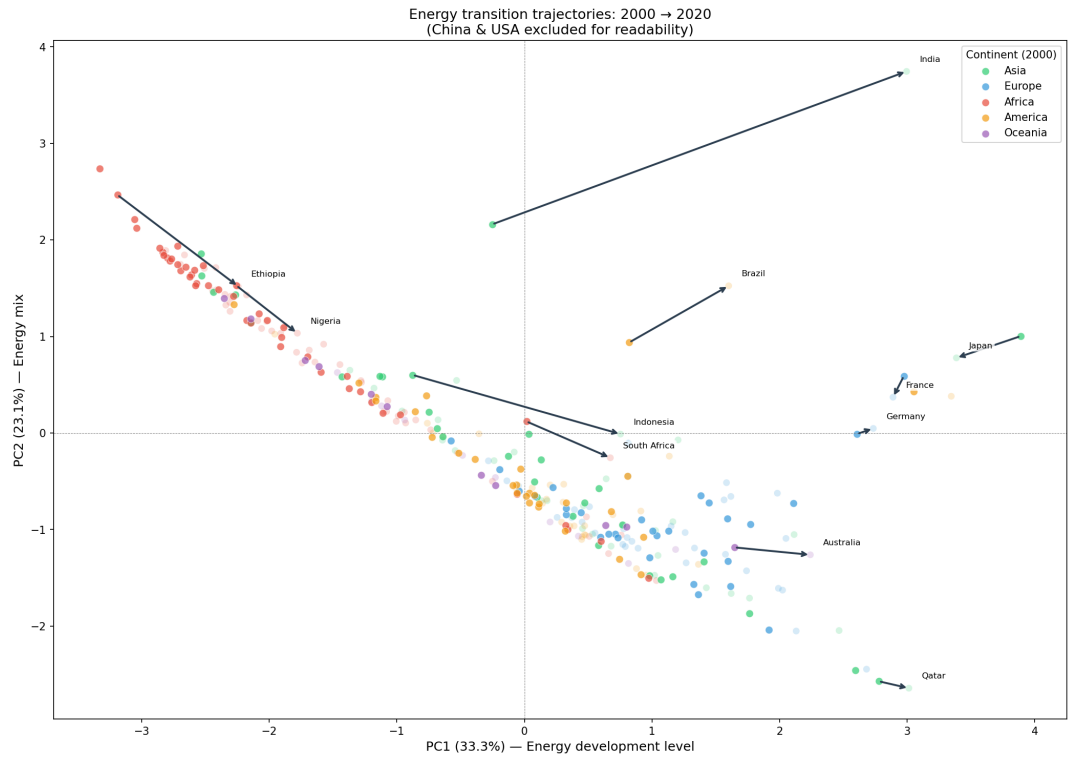


Figure 18: Energy transition trajectories for selected countries (2000 → 2020). Colored points are 2000 positions; the faded background shows 2020 positions. China and USA excluded for readability.